

Complete the sequence: what comes next?

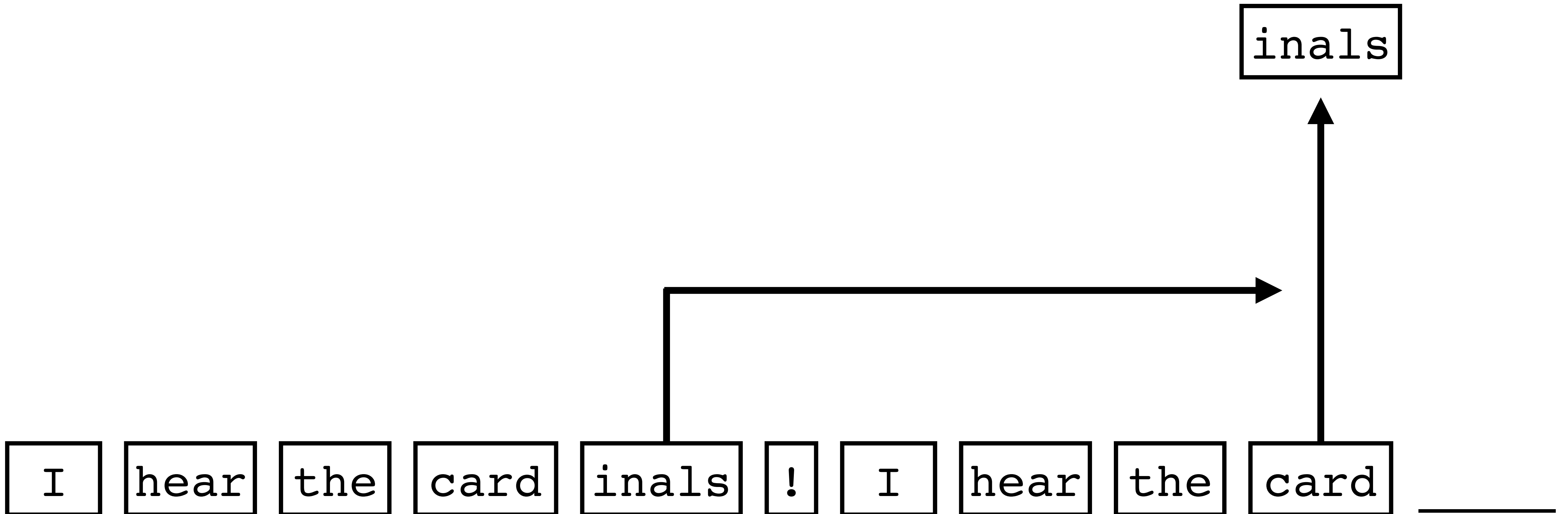
I hear the cardinals ! I hear the card _____

Complete the sequence: what comes next?

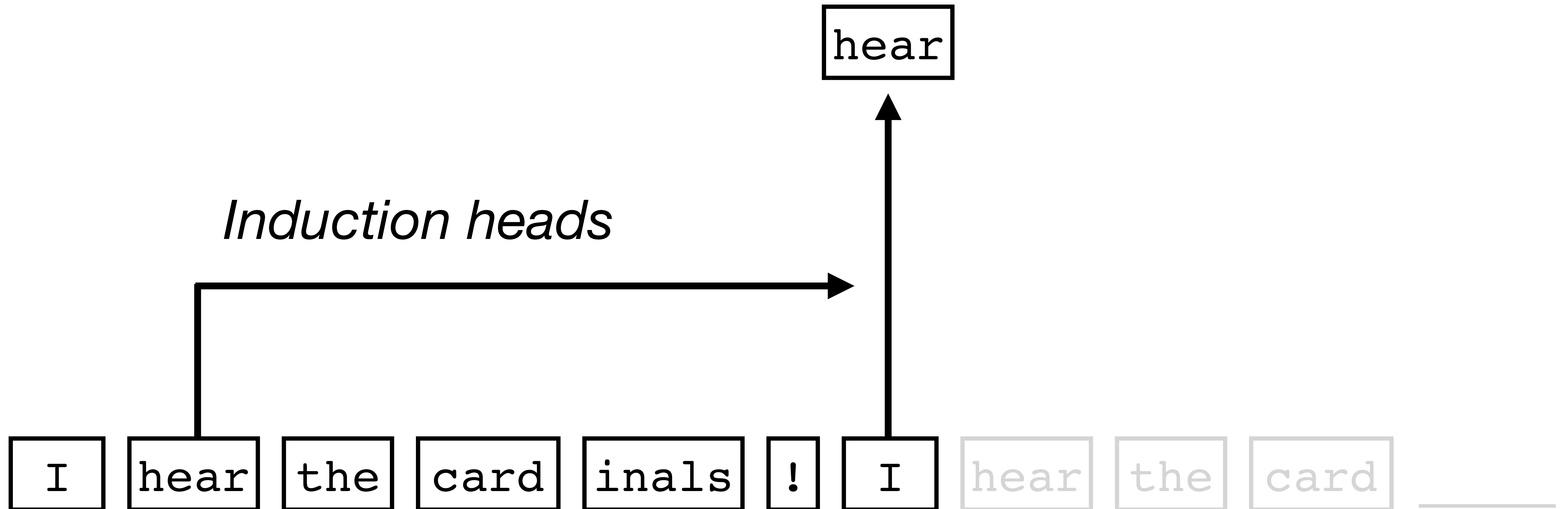
Probably “inals”, right?

I hear the card inals ! I hear the card _____

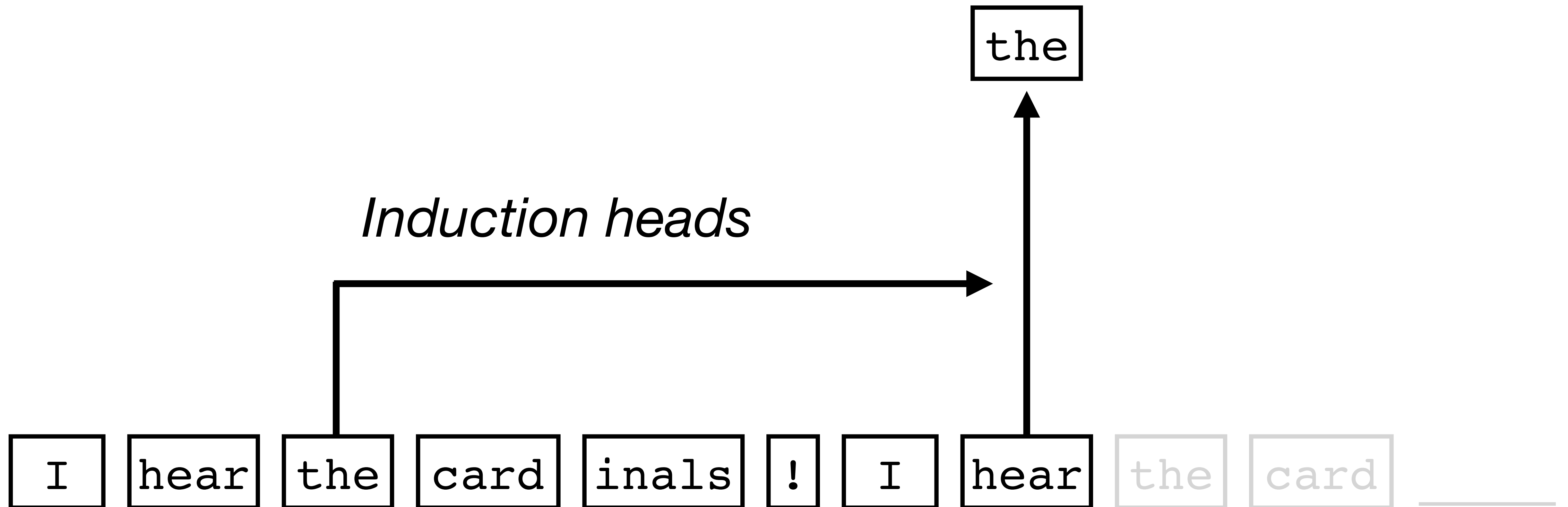
LLMs are really good at completing these sequences.



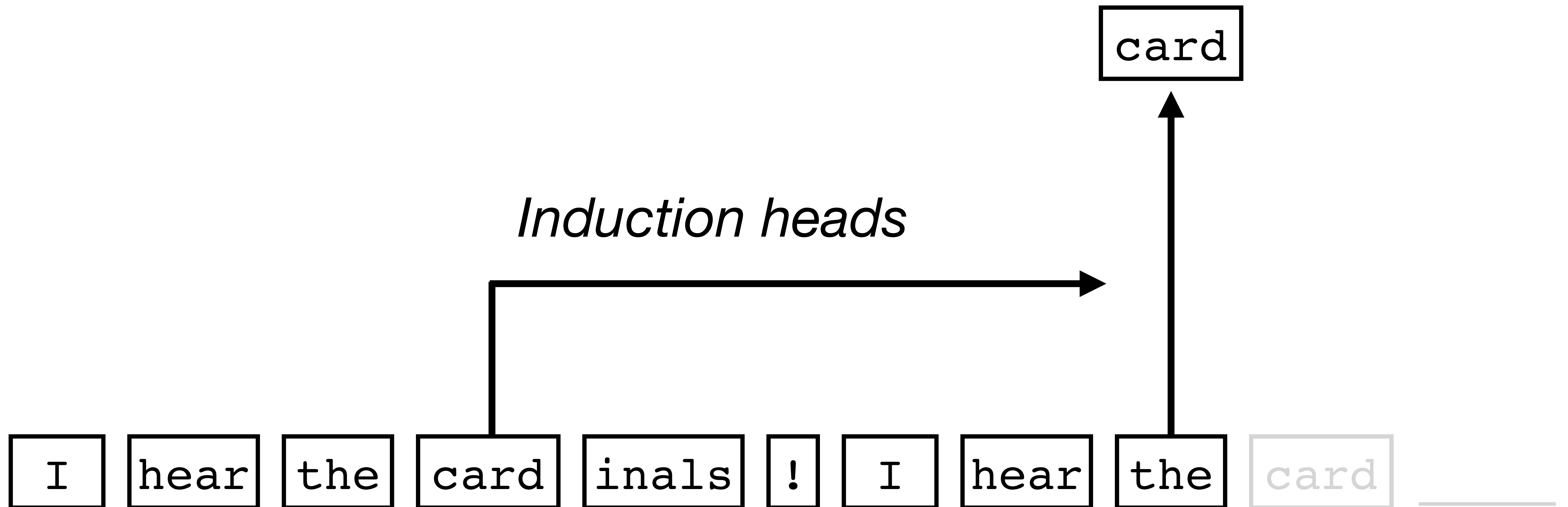
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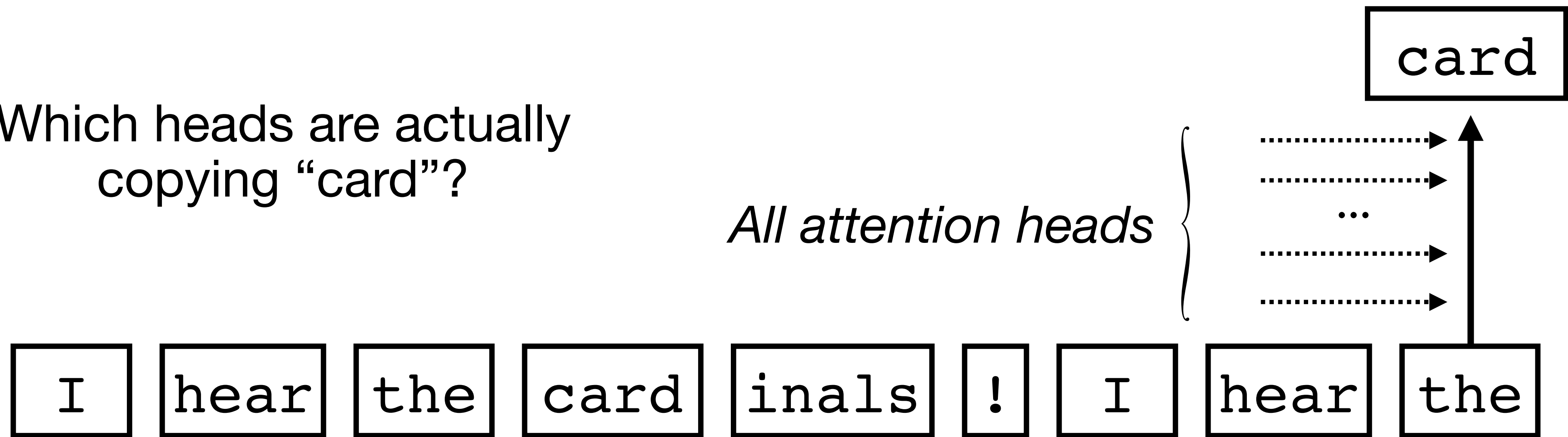


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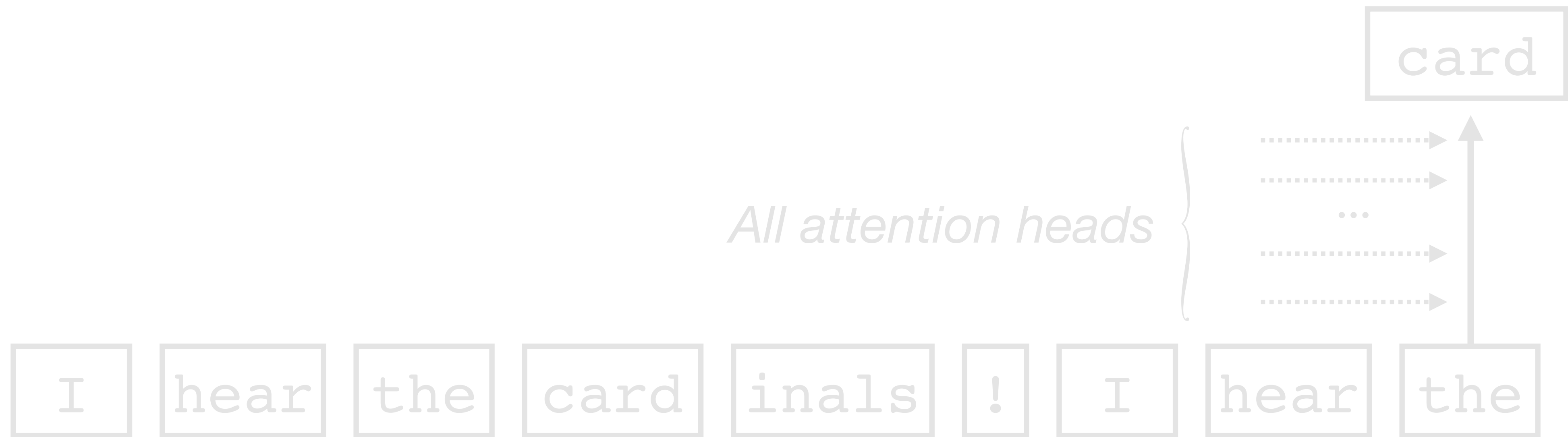


How do we find attention heads that copy?

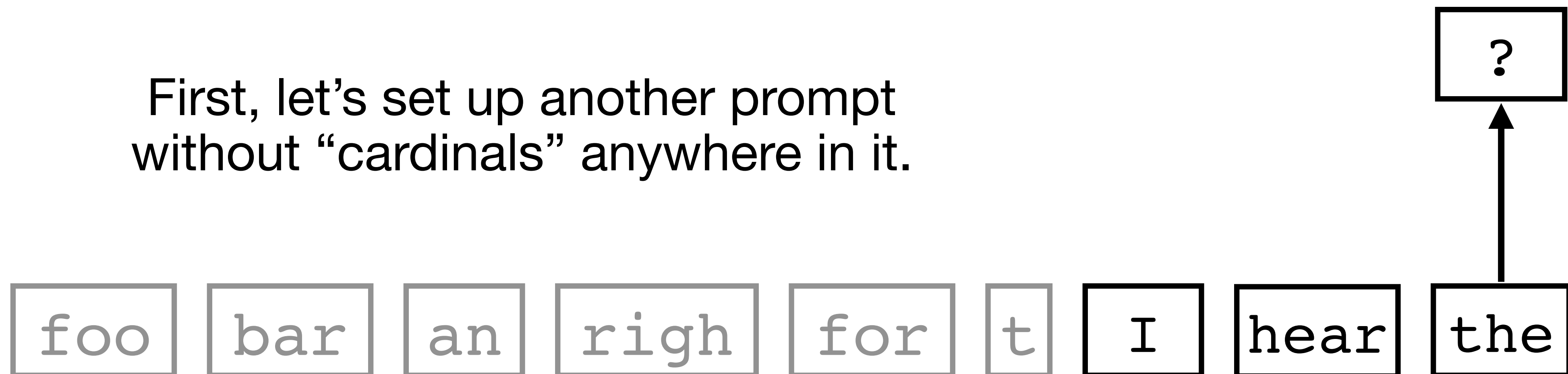
Which heads are actually copying “card”?



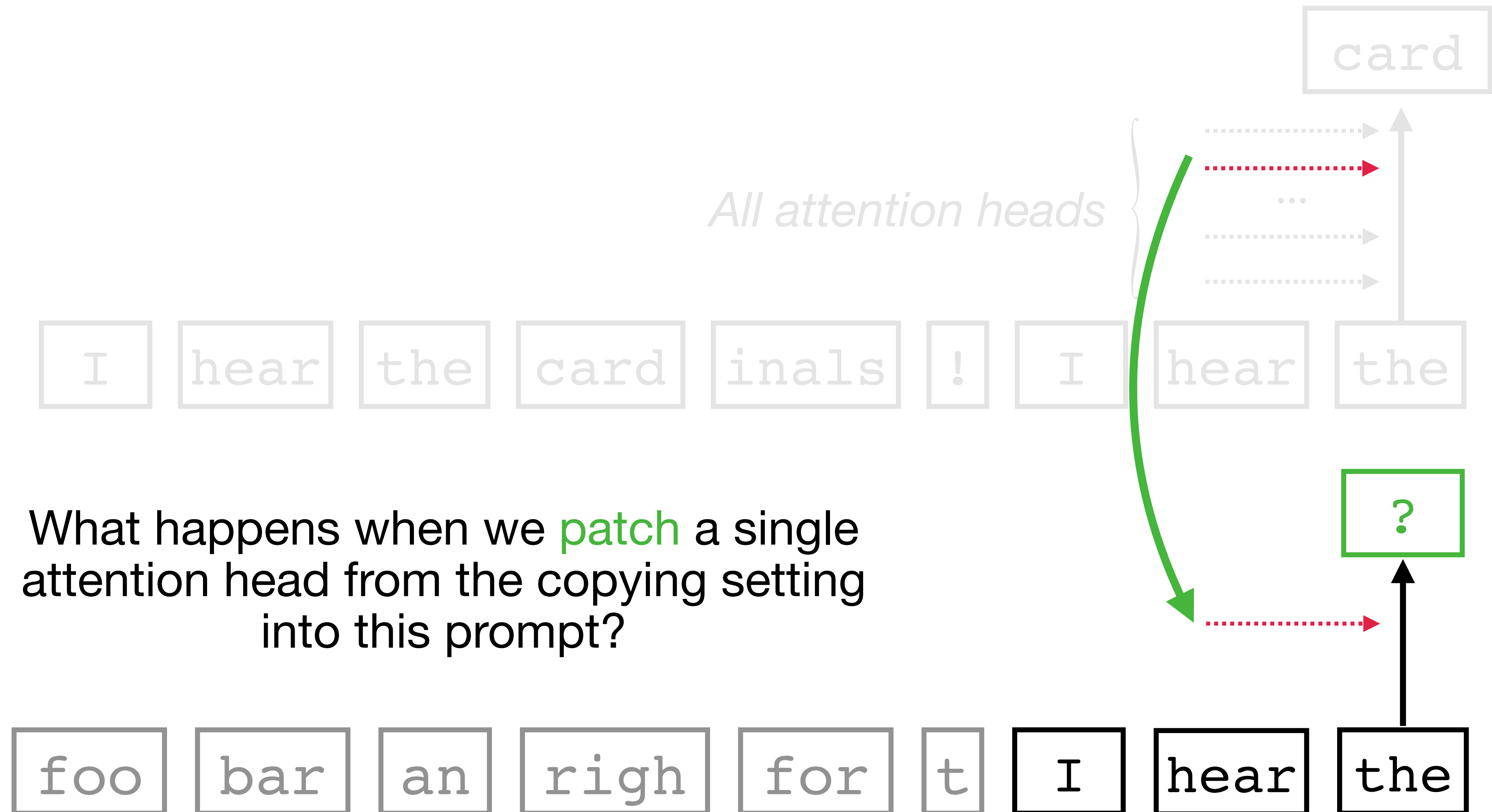
How do we find attention heads that copy?



First, let's set up another prompt without "cardinals" anywhere in it.



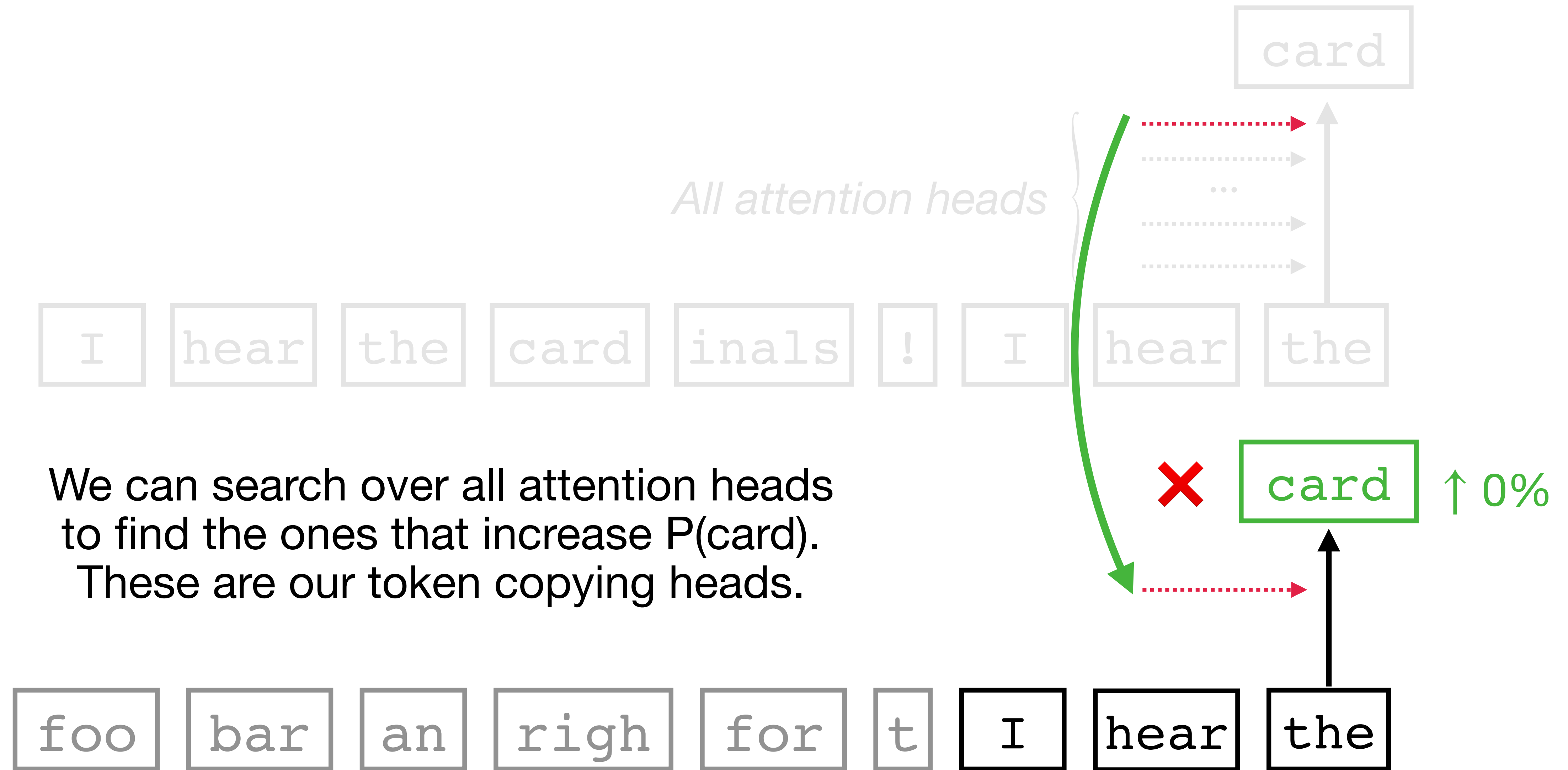
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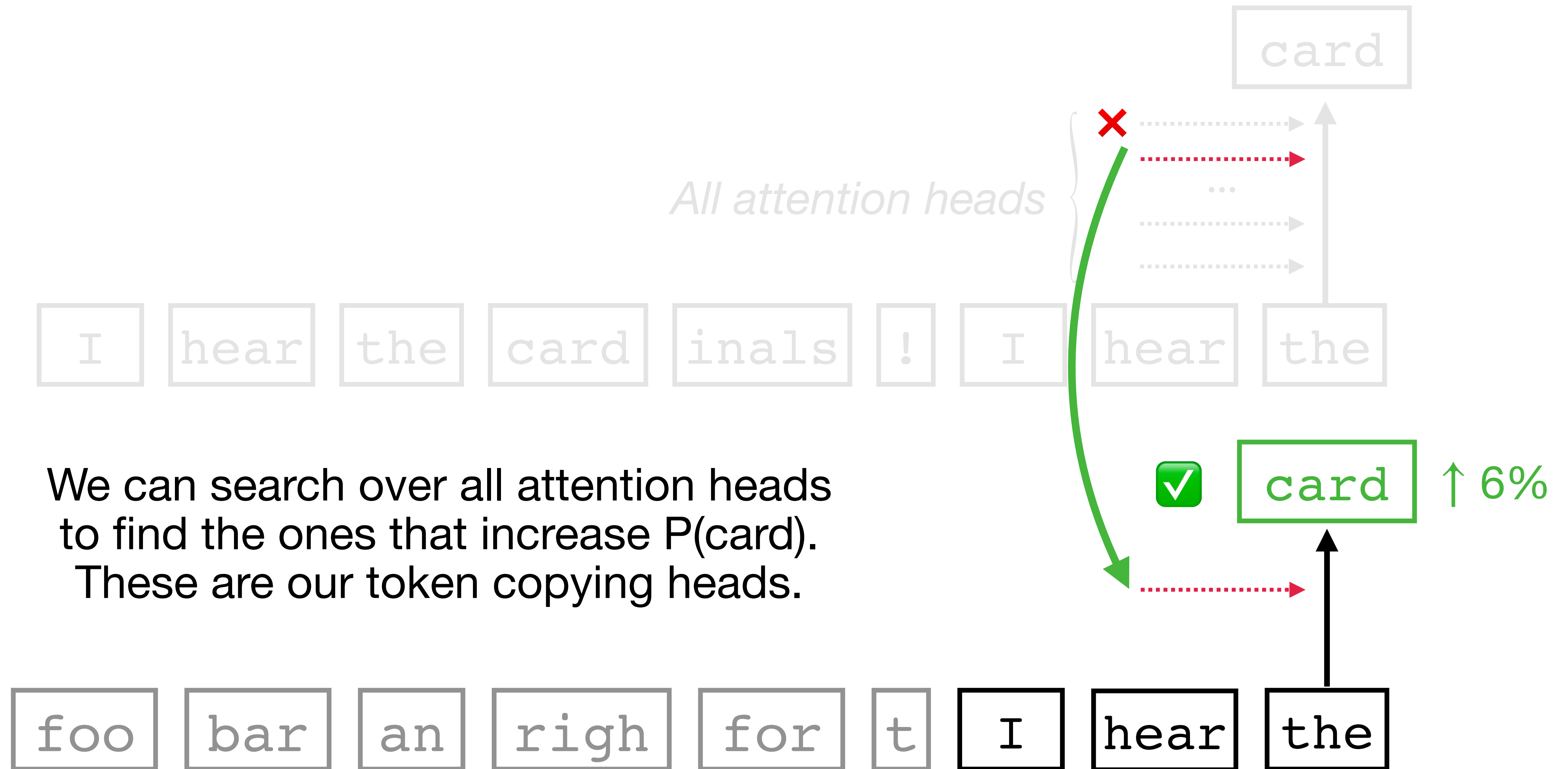
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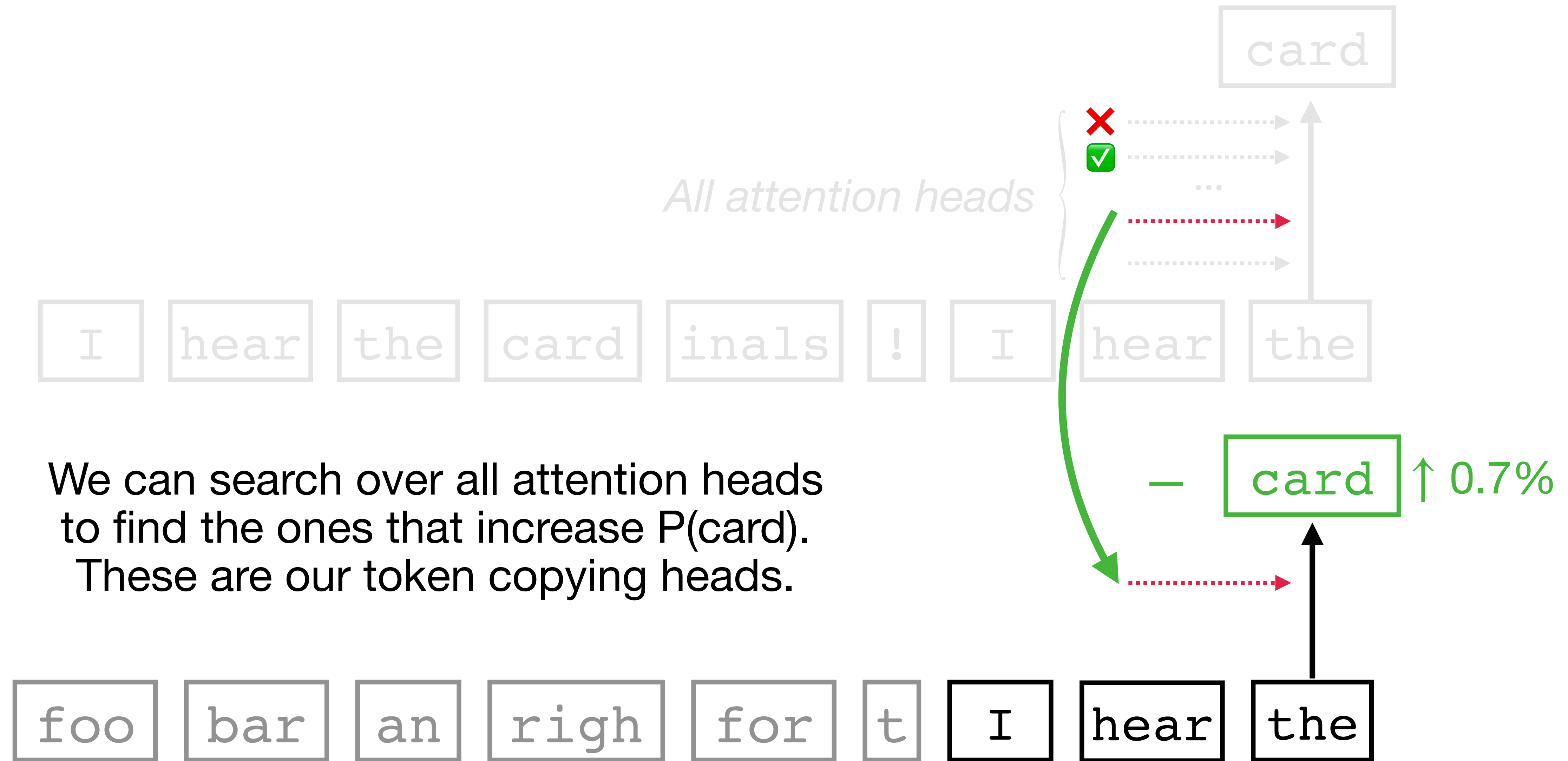
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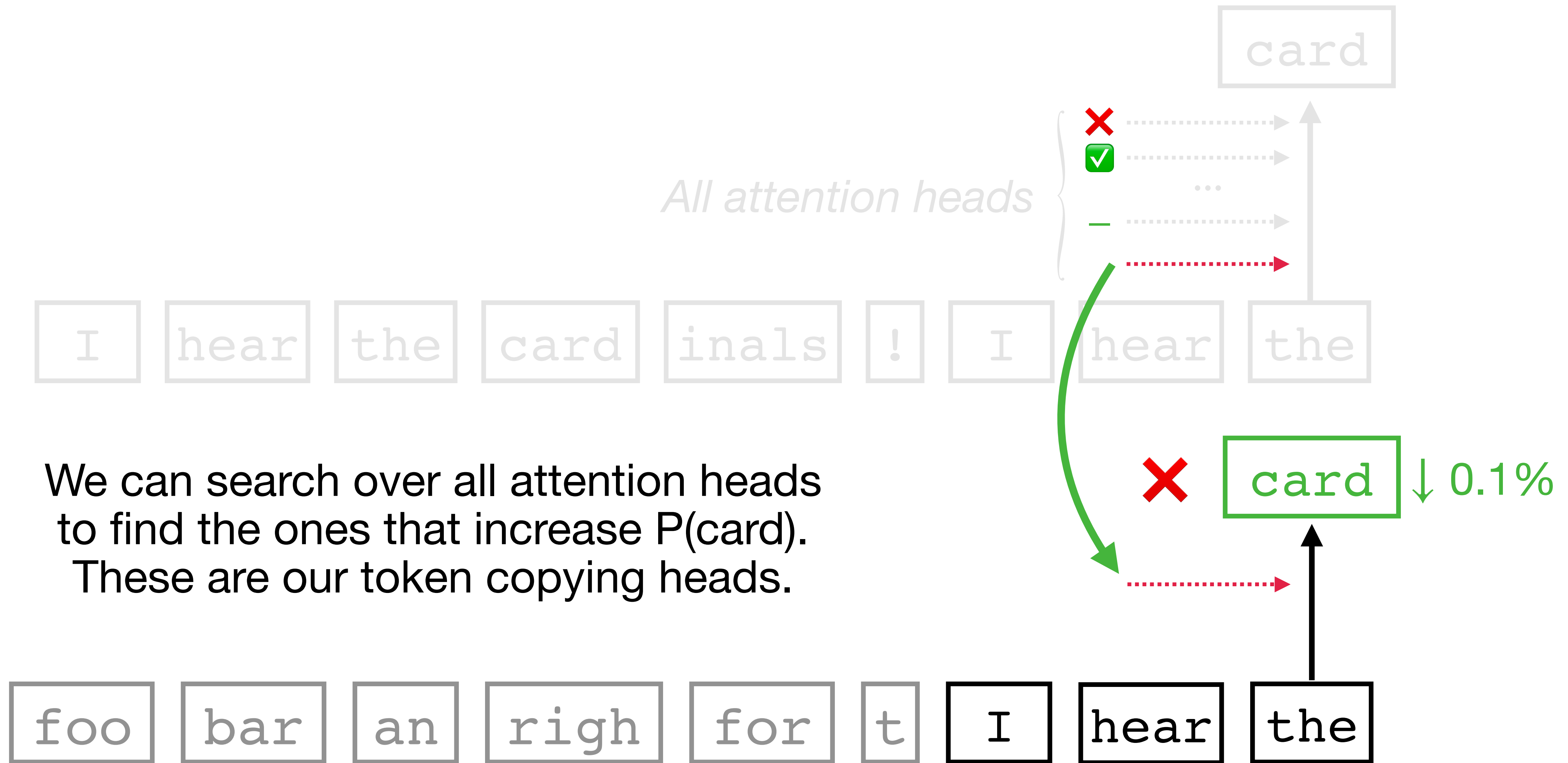
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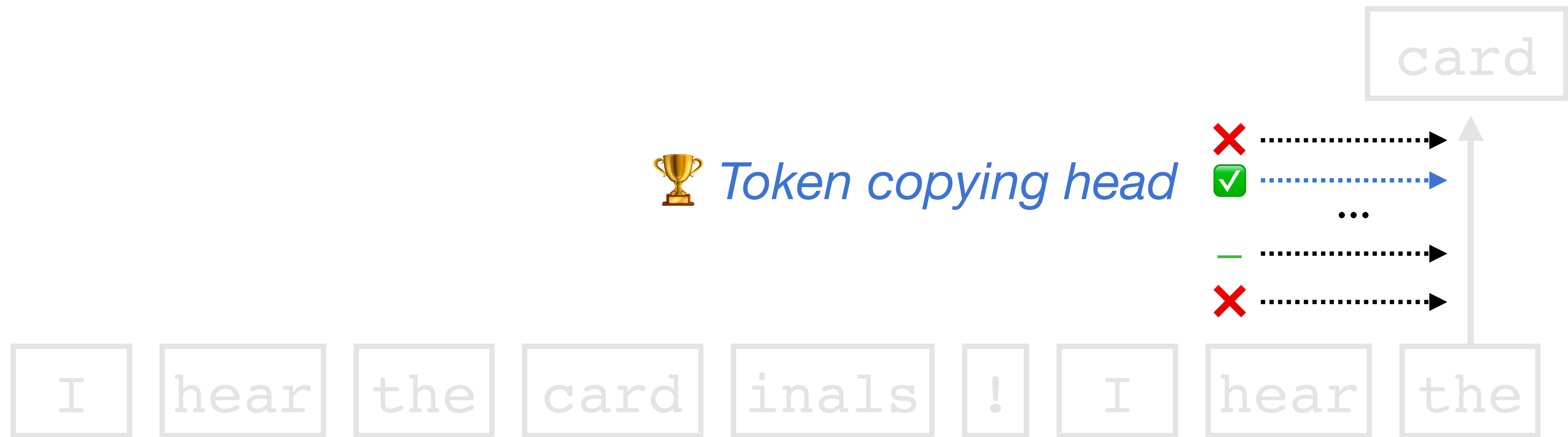
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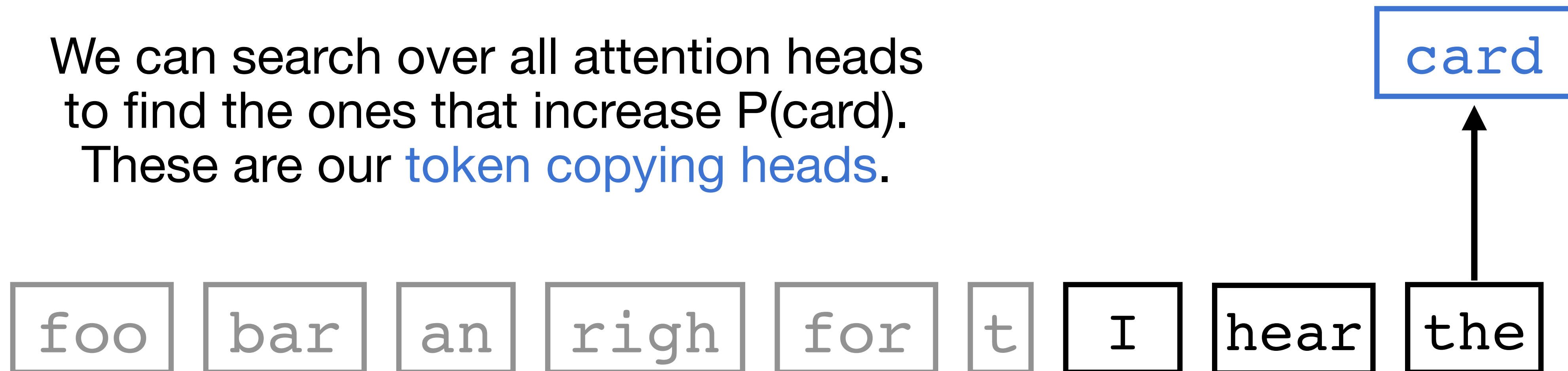
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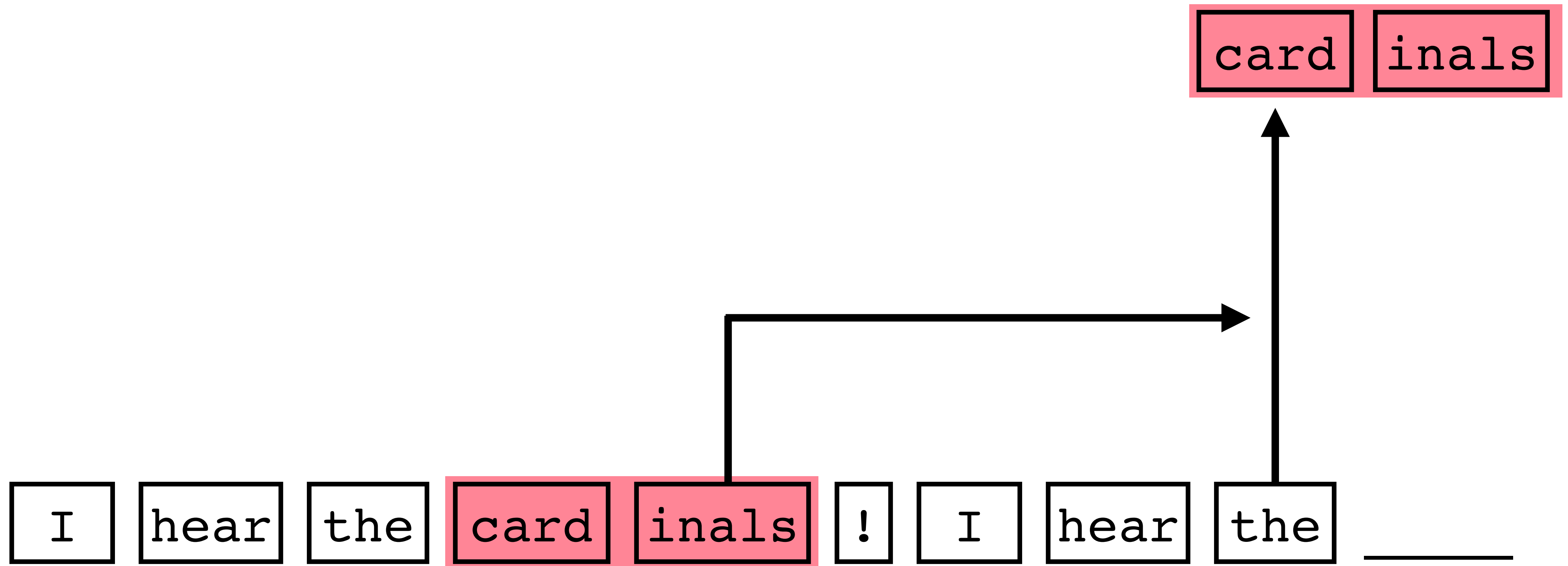
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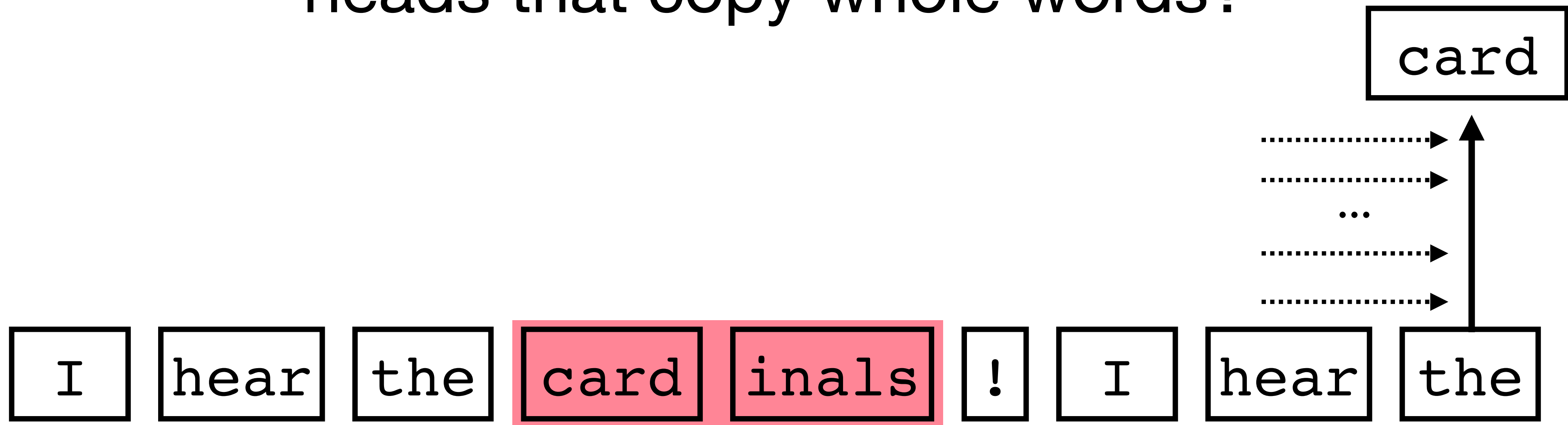
We can search over all attention heads to find the ones that increase $P(\text{card})$.
These are our **token copying heads**.



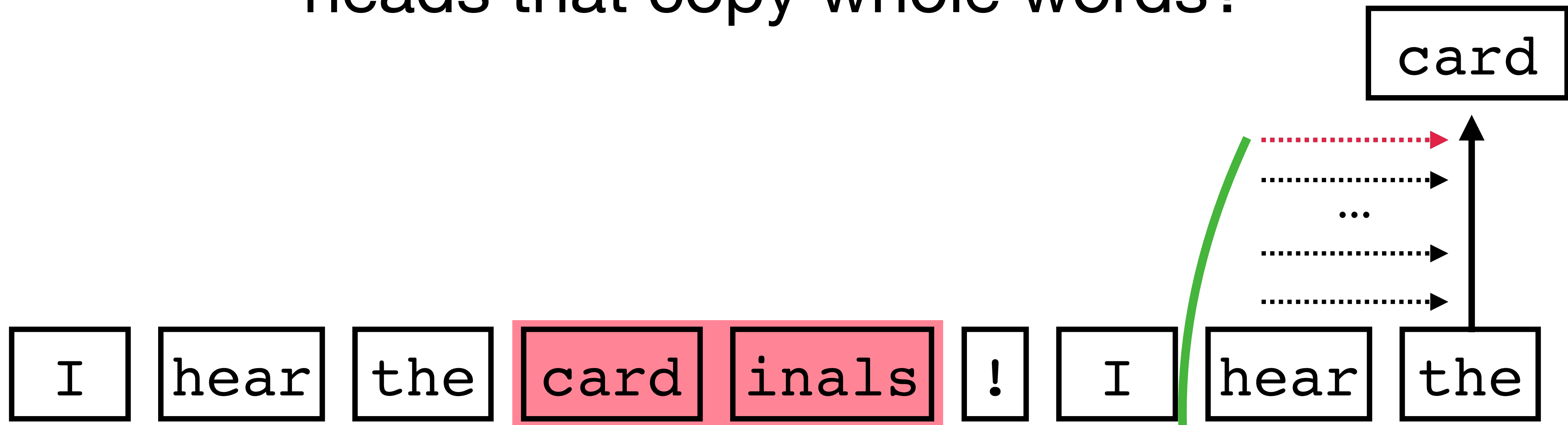
But what if, instead of copying one token at a time, the LLM copies entire *words* at a time?



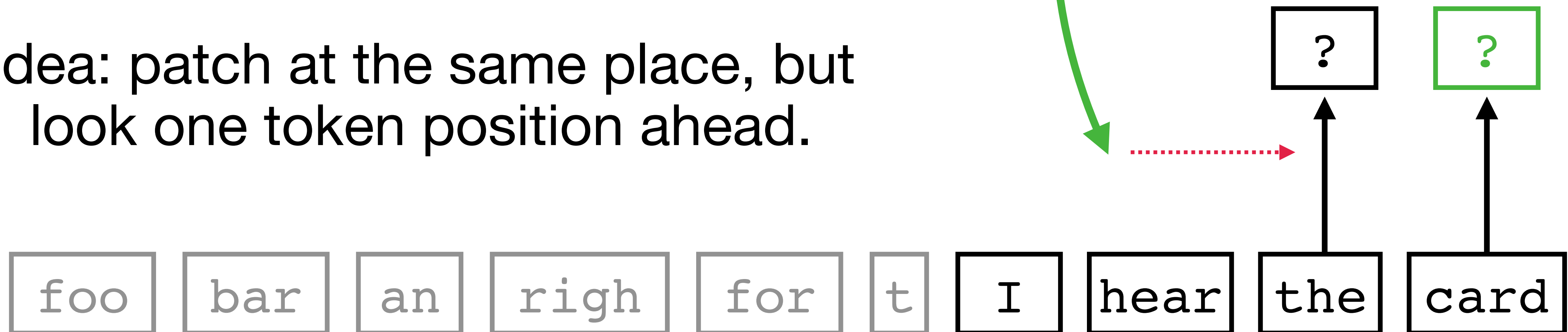
Using patching, how could we find heads that copy whole words?



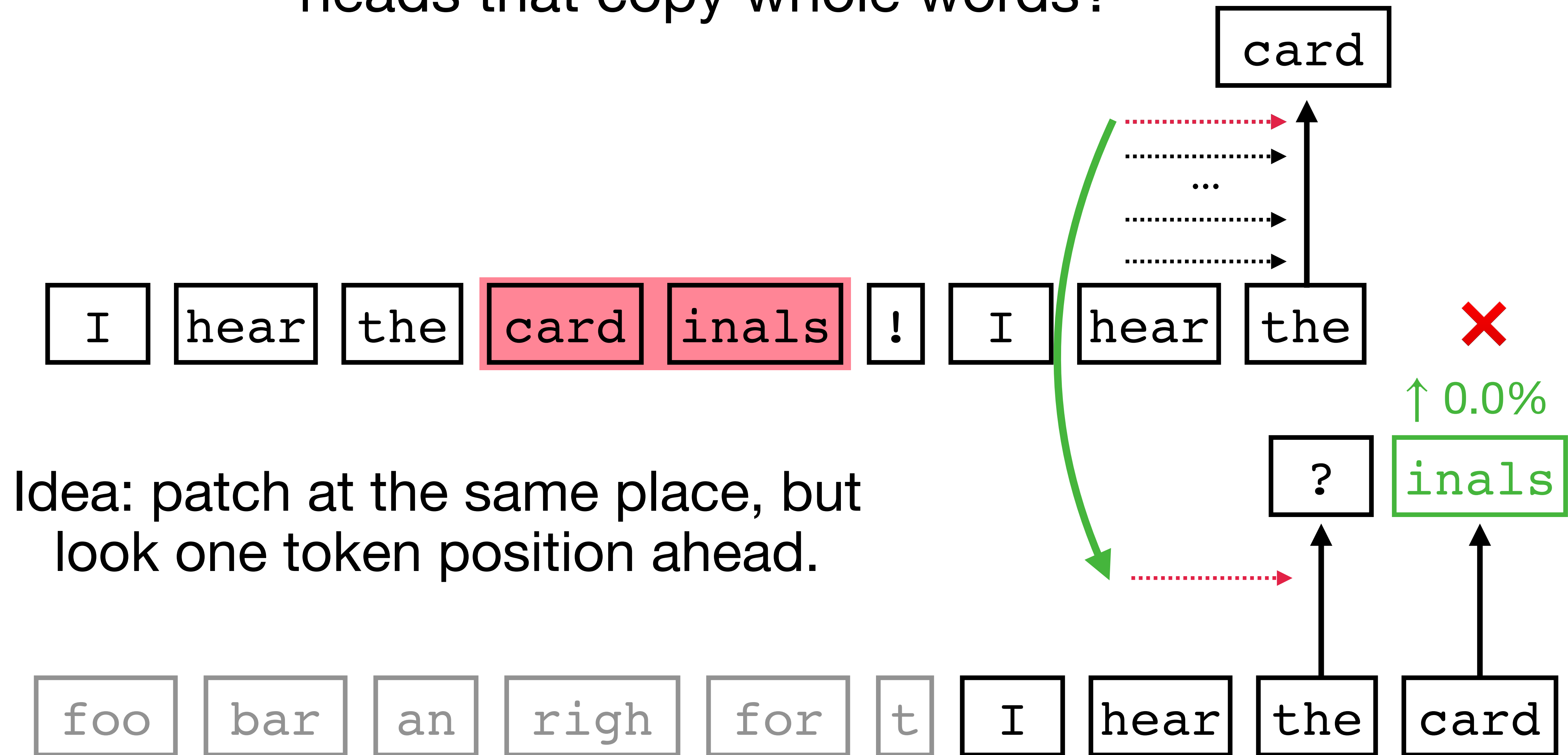
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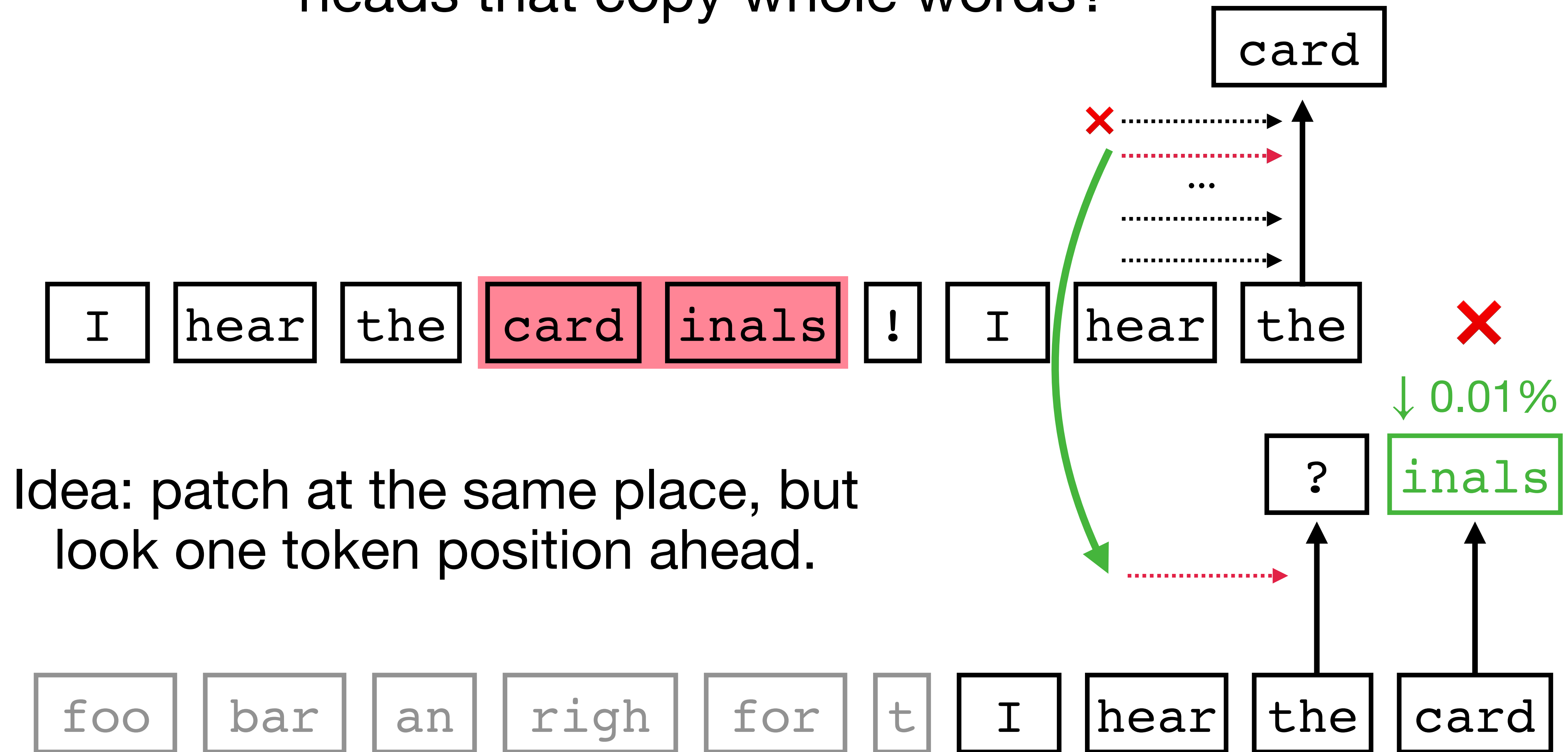
Idea: patch at the same place, but look one token position ahead.



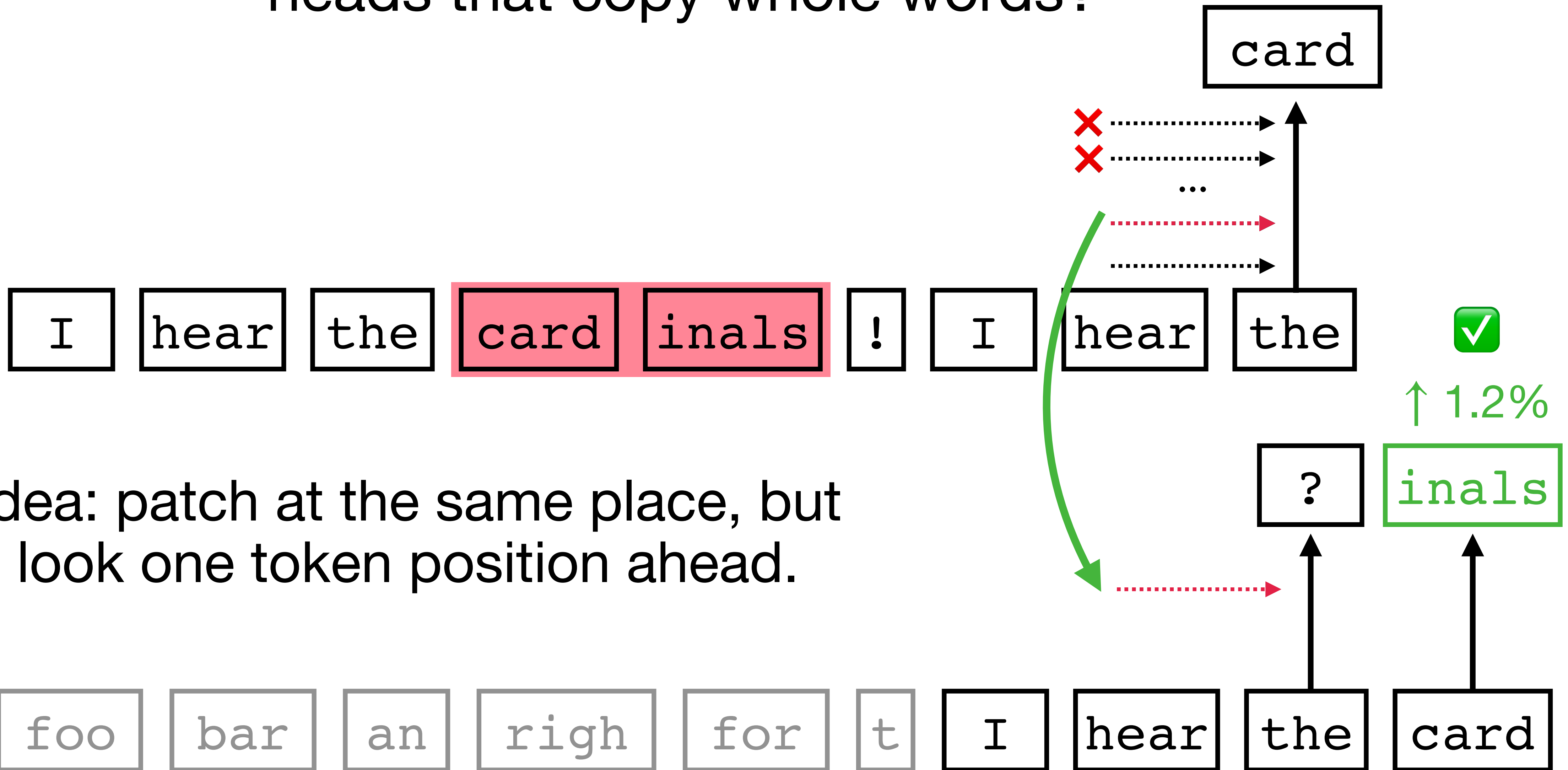
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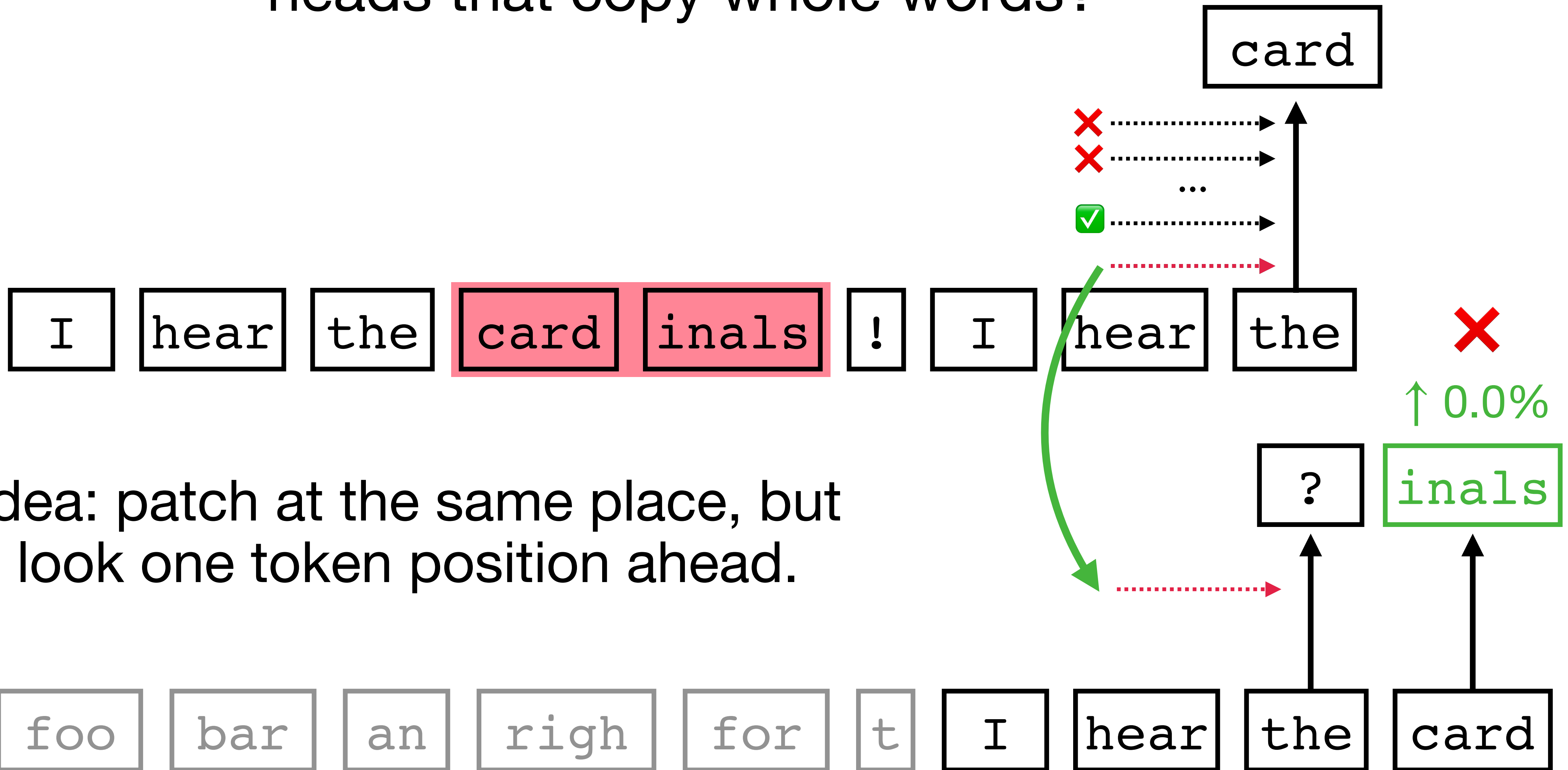
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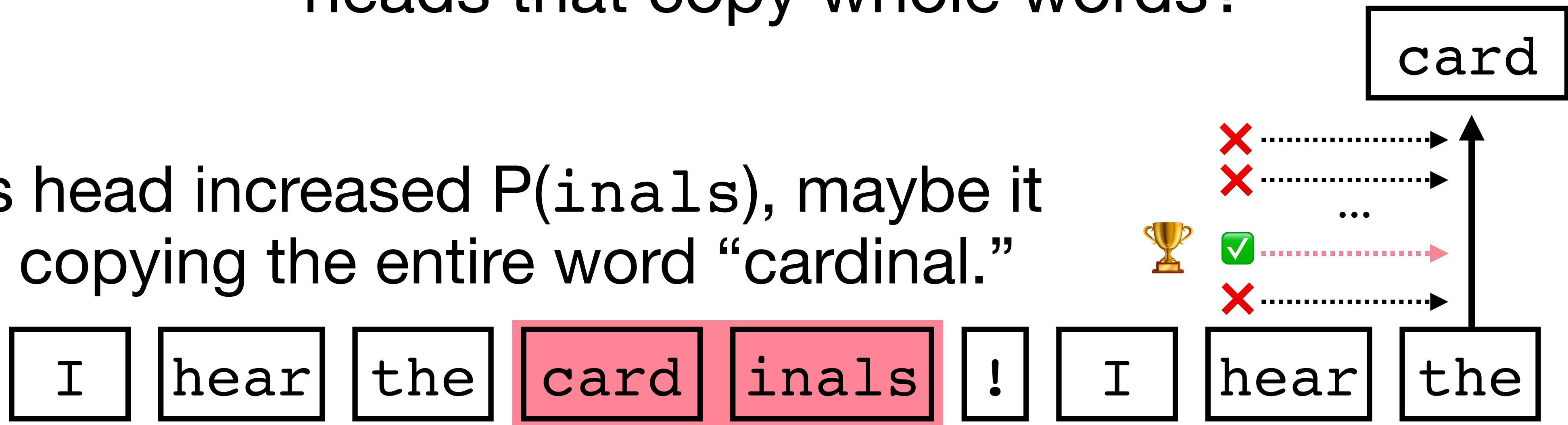


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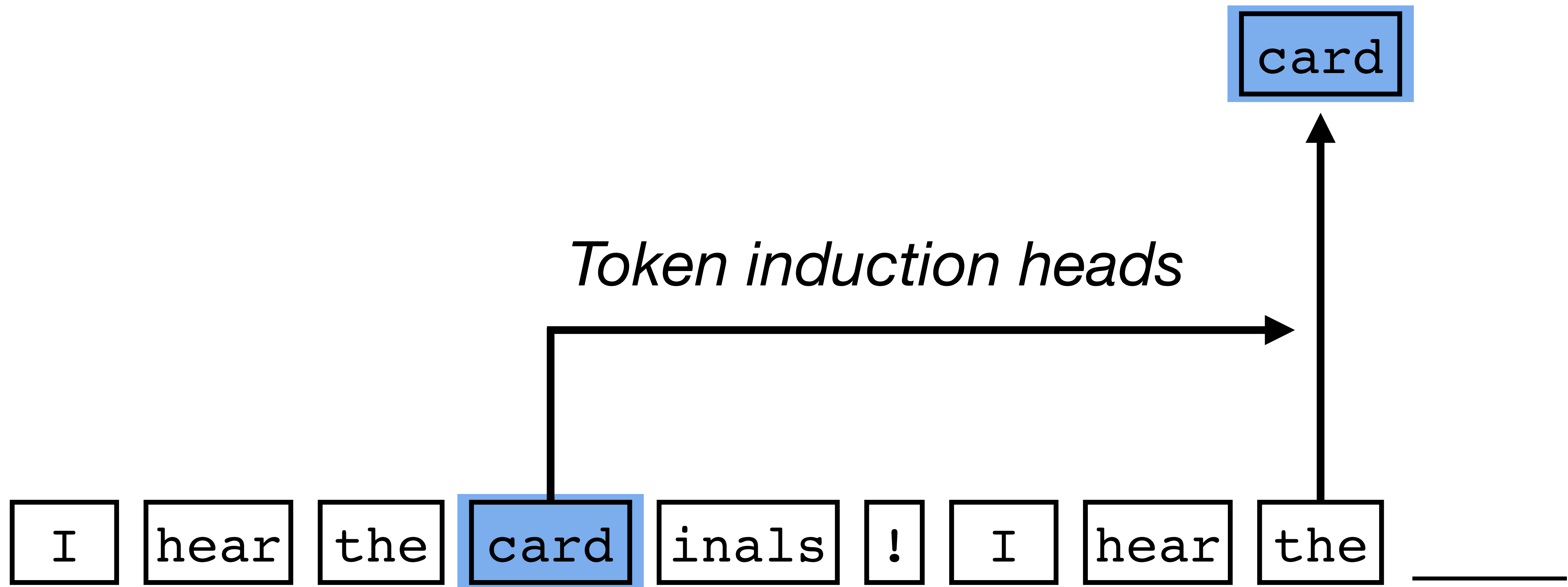


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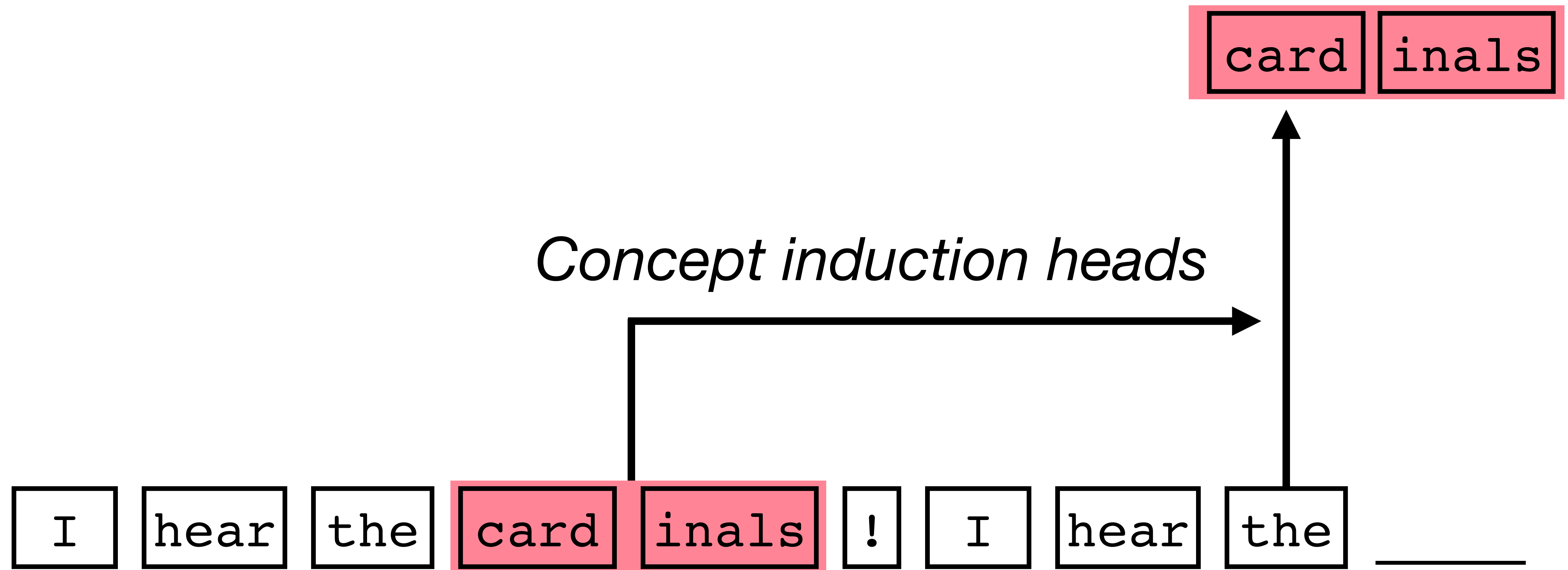
If this head increased $P(\text{inals})$, maybe it was copying the entire word “cardinal.”



So now we have one set of heads that increases $P(\text{card})$ at the next token...

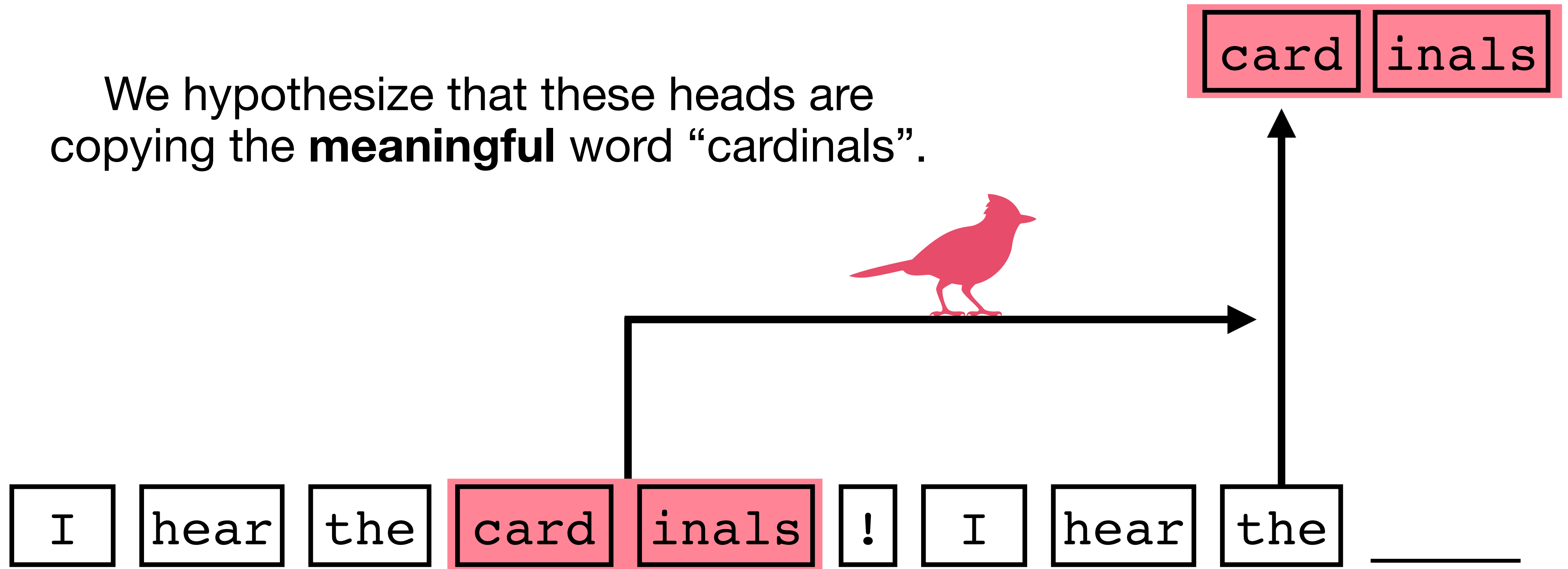


And another, separate set of heads that increases $P(\text{inals})$ at the next-next token.

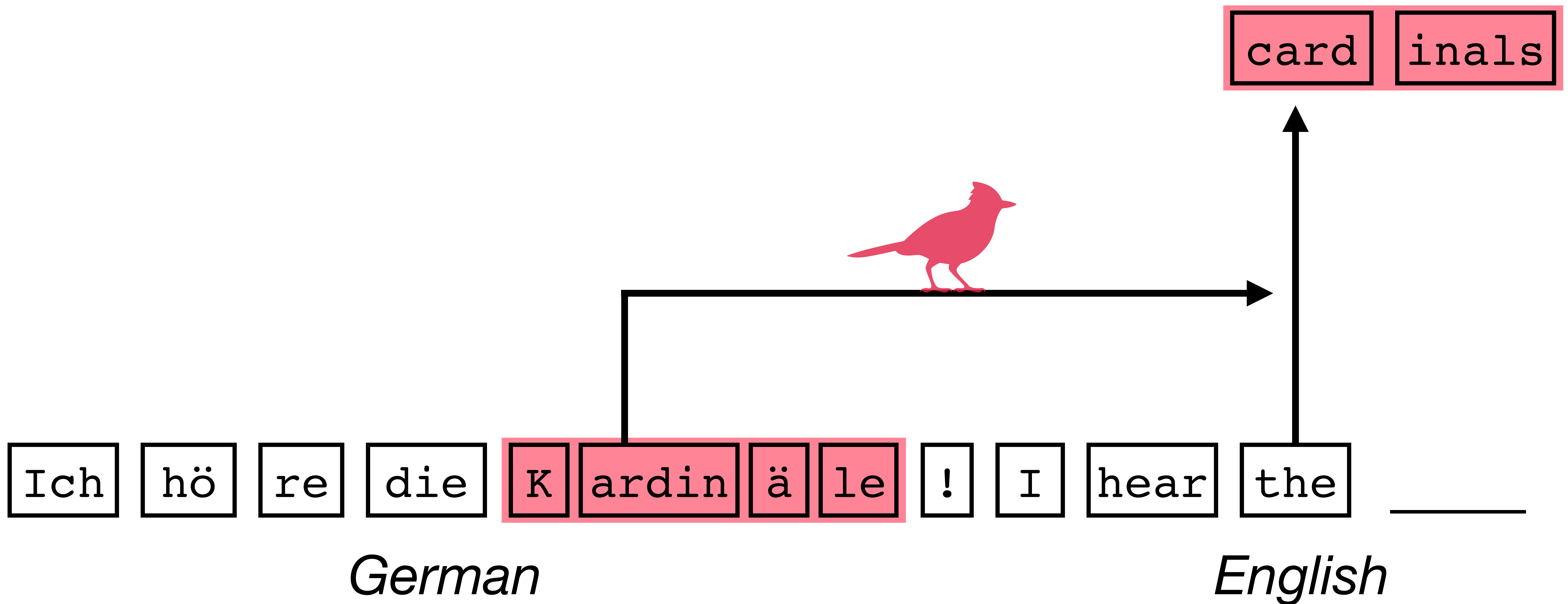


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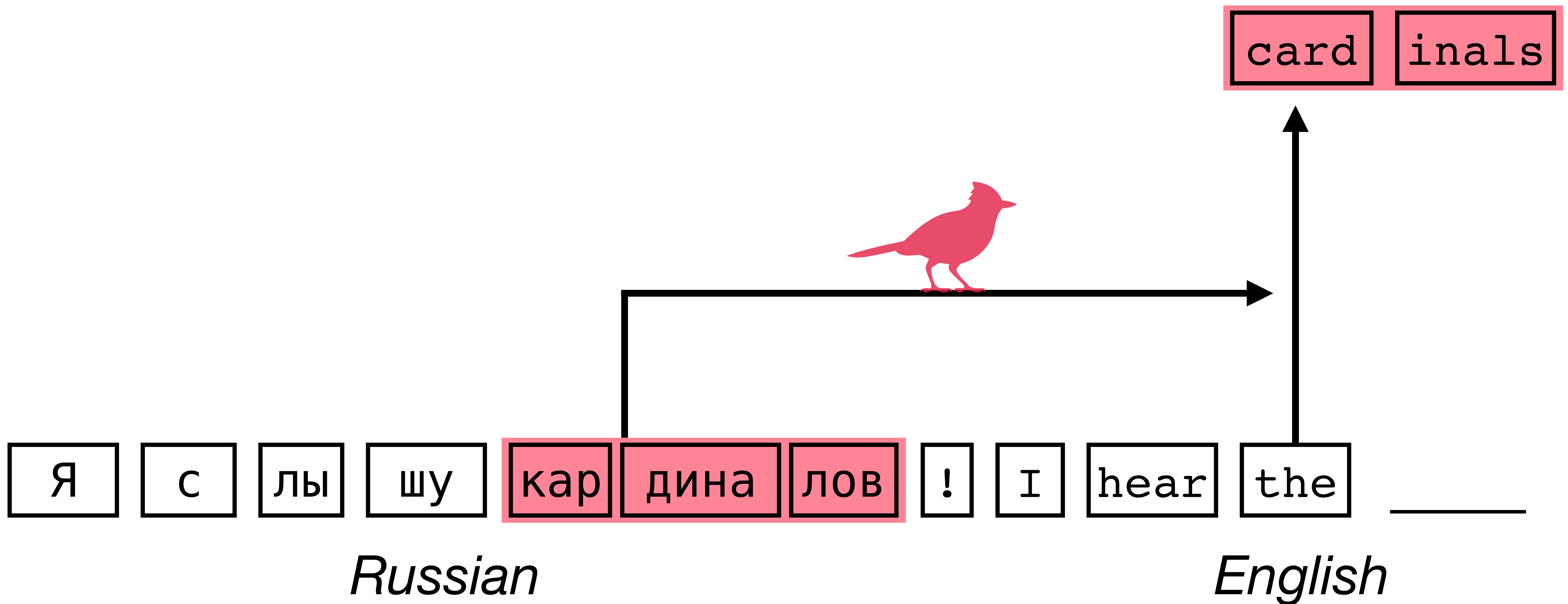
We hypothesize that these heads are copying the **meaningful** word “cardinals”.



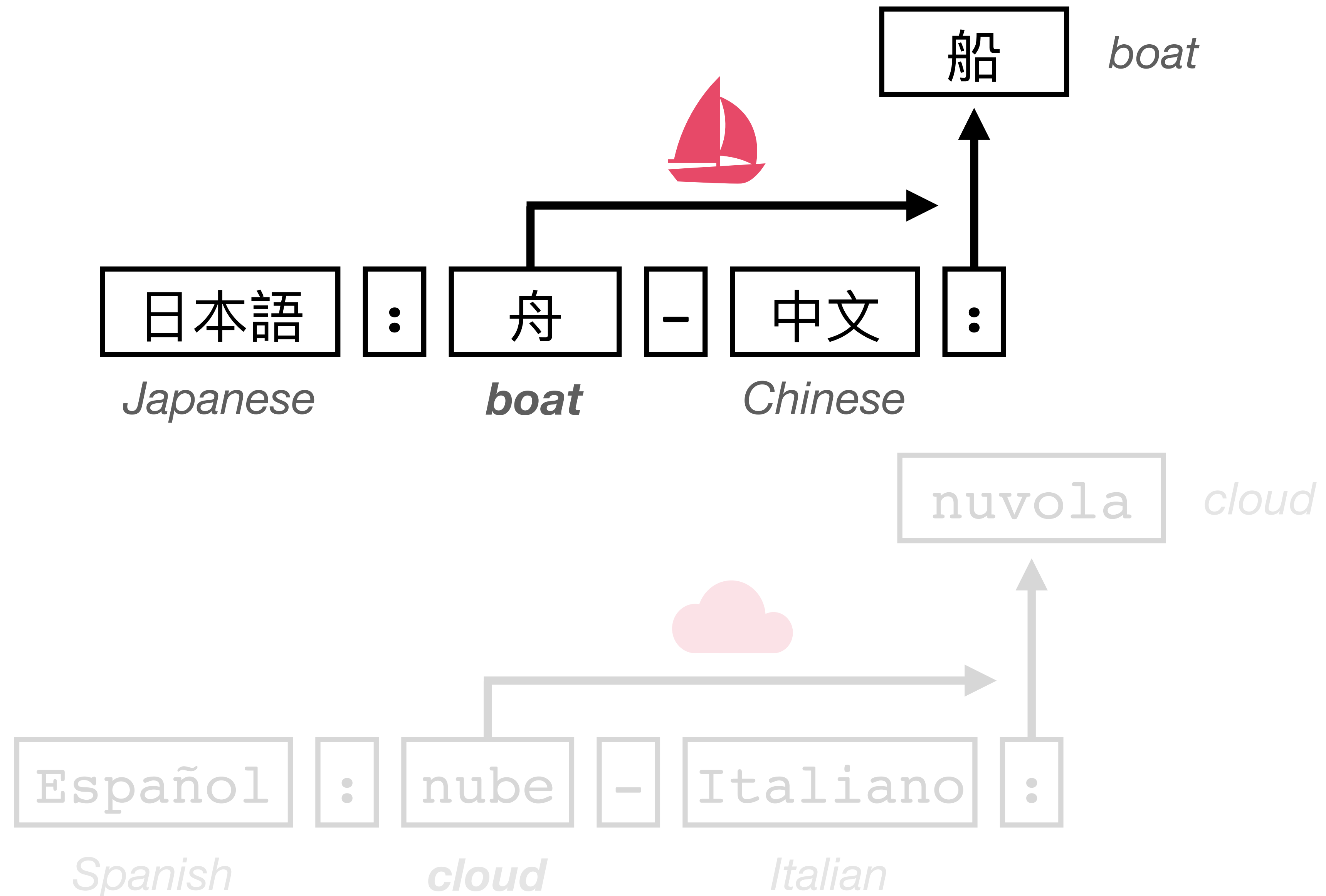
If concept induction heads copy word **meanings**, they should function the same regardless of how a word is written.



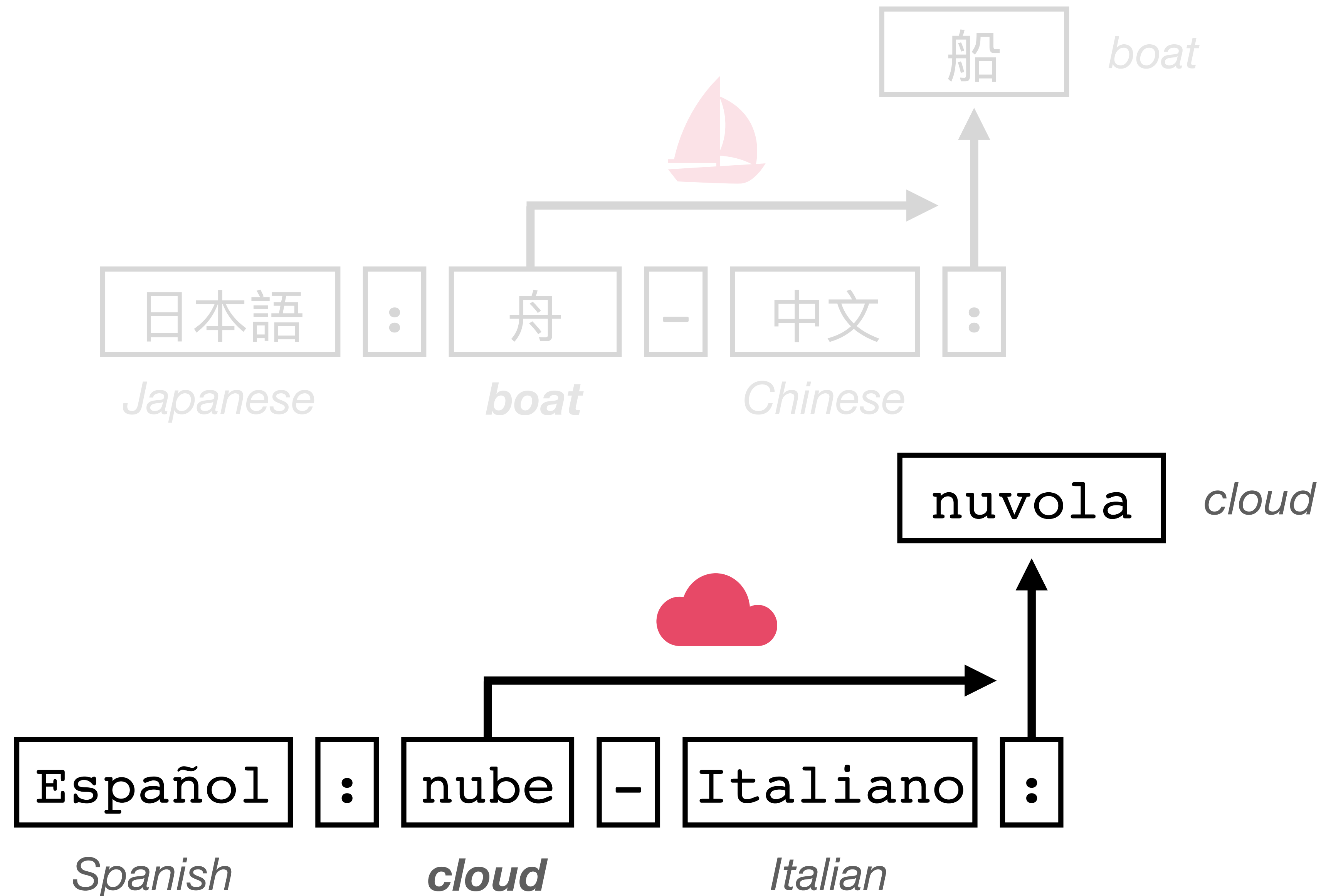
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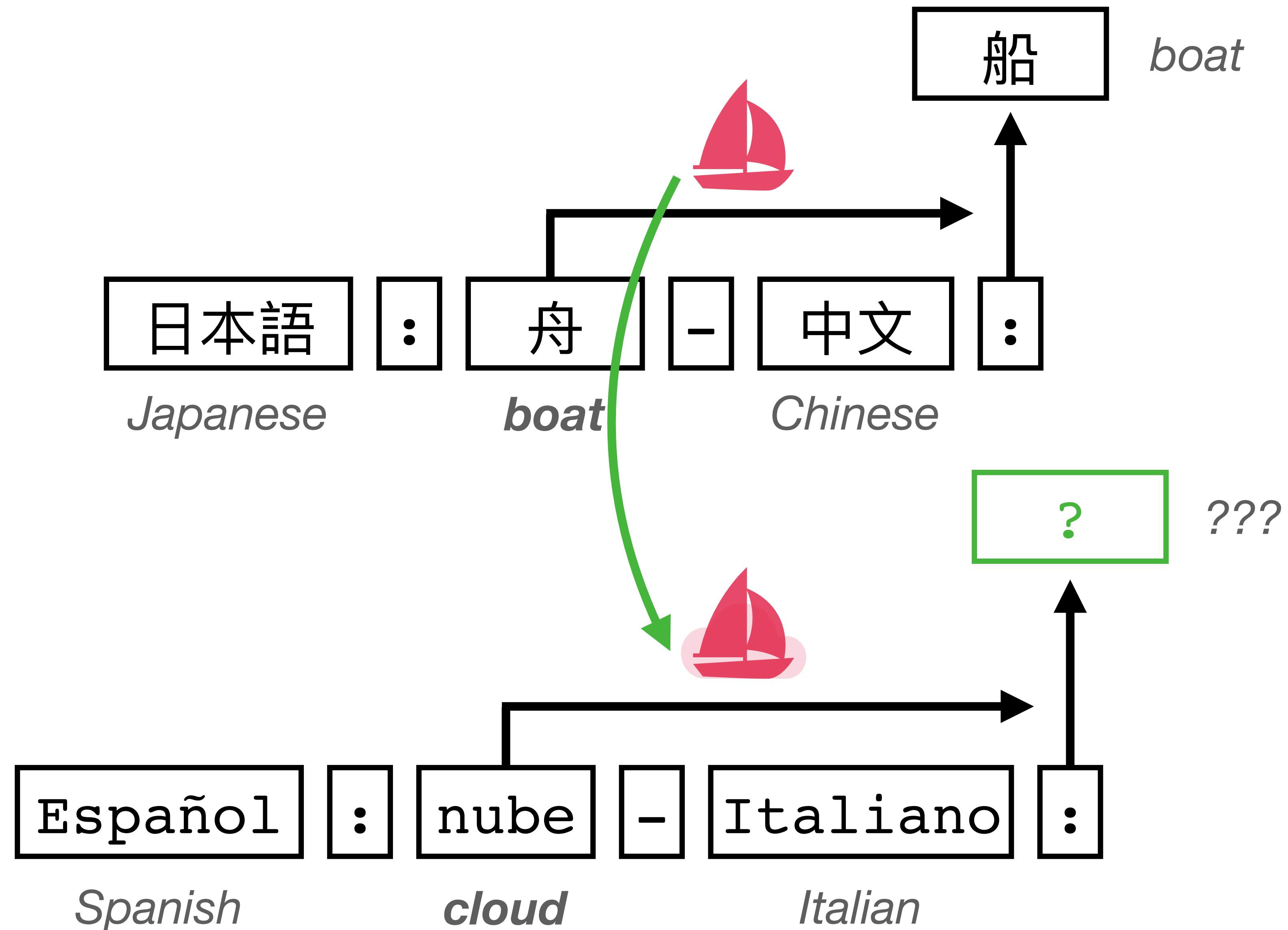
We find this is true! They are used to translate words.



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What happens if we patch these heads into a new context?



It causes the model to output “boat” in Italian!

